

Biomass Gasification

Project Report submitted by

Kushal and Hardhik

R.kushal vardhan reddy(112301026)

P.Hardhik(132301024)

Guided by

Dr. krishna sesha giri



INDIAN INSTITUTE
OF TECHNOLOGY
PALAKKAD

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Abstract :

In biomass downdraft gasification , Gasifier converts solid biomass into useful gaseous fuel, syngas. Gasification of biomass is initiated with the heating of biomass together with limited air. There are four stages: drying, pyrolysis, oxidation, and reduction. Downdraft gasifiers are widely preferred because they produce cleaner gas with lower tar compared to other designs.

The aim of this work is to observe the behavior of the gasifier during operation, including how the temperature changes inside different reaction zones and what gases are produced. During the experiment, biomass will be loaded into the reactor, air will be supplied at a controlled flow rate, and after the beginning of steady operation, temperature and gas composition are measured. Important components of the gas, like CO, CO₂, H₂, CH₄, O₂, and N₂, are recorded.

The present study will help in understanding the efficiency, stability, and practical usefulness of downdraft gasification. The experimental results are compared with values reported in well-known global references. On the whole, the project gives a very clear understanding of the gasification process and establishes that the downdraft gasifiers are suitable for small-scale, clean energy applications.

Introduction :

Organic materials like wood fragments, sawdust, coconut shells, agricultural waste, and other plant-based residues are referred to as biomass. These materials are renewable, widely available, and can be used as a sustainable source of energy.

Biomass gasification is a thermochemical process in which solid biomass is converted into a combustible gas mixture called syngas. This happens when biomass is heated with a limited amount of air, not full combustion.

The gasification process occurs through four major stages:

1. Drying – removal of moisture
2. Pyrolysis – decomposition of biomass releasing volatiles
3. Oxidation – partial combustion producing heat
4. Reduction – conversion of char into CO, H₂, and CH₄

The resulting syngas contains CO, H₂, CH₄, CO₂, O₂, and NO.

Types of biomass Gasifiers

Biomass gasifiers are mainly classified into:

1. Updraft Gasifier

- Air enters from the bottom and gas flows upward
- High tar content
- Simple structure but not suitable for engine applications

2. Downdraft Gasifier

- Air and gases move downward through hot oxidation zone
- Naturally cracks tar
- Produces **low-tar syngas** suitable for engines

3. Cross-draft Gasifier

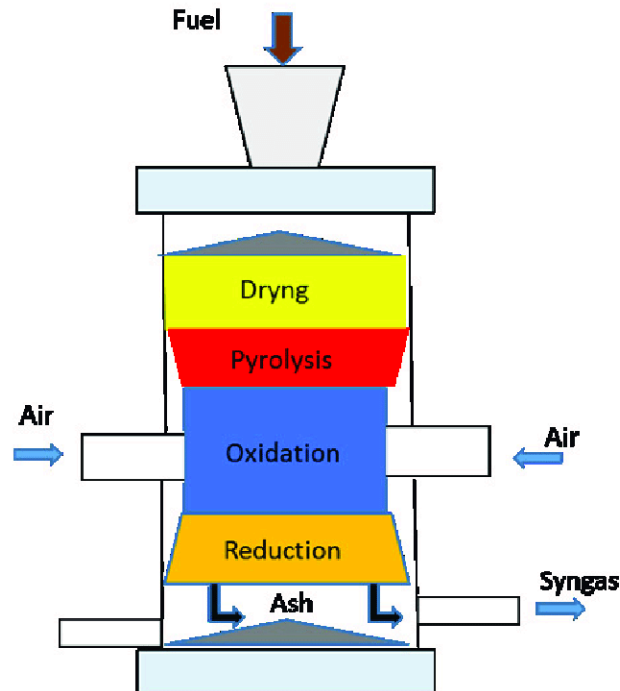
- Air moves horizontally
- High temperature but sensitive to fuel quality

4. Fluidized Bed Gasifier

- Sand/bed material fluidized by air
- Suitable for large industrial applications

Why Downdraft Gasifier ?

- They naturally produce low-tar syngas due to tar vapours passing through the high-temperature oxidation zone.
- They are simple, reliable, and easy to operate for small-scale systems.
- Suitable for rural electrification, heating, and lab experiments.
- Practical implementations such as the IISc Hosahalli village gasifier show durability, clean operation, and low maintenance.
- The composition of syngas (CO 18–25%, H₂ 10–20%, CH₄ 1–5%) is suitable for small power applications.



Biomass Downdraft Gasifier

Motivation :

1. Abundant Availability of biomass

- One of the most accessible renewable resources in our environment is biomass.
- Massive volumes of organic materials, wood waste, and agricultural residues are thrown away every day.
- Instead of being dumped or burned, this biomass can be used as a valuable energy source.

2. Cutting Down on Environmental Contamination

- Gasification reduces hazardous emissions by turning this waste into clean, usable fuel.
- This promotes cleaner environments and improved waste management.

3. Need for Clean and Sustainable Energy

- Biomass gasification provides a renewable, carbon-friendly solution that can replace a portion of fossil fuel use.

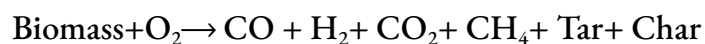


Literature Review :

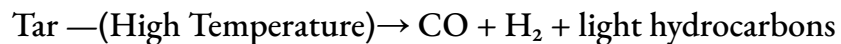
Working principle :

Downdraft gasification is a thermochemical process that converts solid biomass into a combustible gas mixture by reacting it with a limited supply of air . The fuel and gases move downward . passing through drying ,pyrolysis , oxidation and reduction zones. The pyrolysis vapors are forced through the hot oxidation zone(800- 1000 c) , which promotes thermal cracking of tar components .Because of this mechanism , downdraft systems produce cleaner syngas than updraft or open-core reactors.

The overall gasification process is represented by the this general reaction



However , in a downdraft design ,tar is minimized because



Key parameters :

The most important parameter reported across studies in the Equivalence Ratio(ER)

Defined as :

$$\text{ER} = (\text{Actual Air supply}) / (\text{Stoichiometric Air equipment})$$

Experimental work shows that an ER of 0.20 -0.35 gives the best performance.

- Lower ER(<0.2) = incomplete conversion , high tar
- Higher ER(>0.35) = High temperature but lower gas heating values due to excess oxidation

Typical syngas composition at optimal ER:

- CO: 18–22%
- H₂: 12–18%
- CO₂: 10–12%
- CH₄: 1–3%
- O₂ : 0-2%

The quality of the syngas correlates with :

- Temperature peaks in oxidation and reduction zones
- Consistency of feedstock
- Airflow distribution
- Tar cracking efficiency

Feedstock influence :

Feedstock properties heavily affect reactor stability and gas quality. Literature consistently reports optimum performance when biomass has:

- Moisture content: <15–20 %
- Uniform particle size: prevents channeling and maintains even air flow
- Low ash content: reduces slagging and clinker formation

When biomass has high moisture, more heat is consumed in the drying zone to evaporate the water inside the fuel. The energy needed to convert water into vapor is given by:

$$Q_{\text{(evap)}} = M_{\text{(water)}} * H_{\text{(fg)}}$$

Where,

- M (water) = mass of moisture in the biomass
- H(fg) = Latent heat of vaporisation

Lower oxidation temperature means weaker thermal cracking of tar vapours, so more tar escapes unreacted, increasing tar content in the final syngas.

Tar Behaviour in Literature

Tar is one of the main challenges in biomass gasification. Even though downdraft reactors reduce tar significantly, complete elimination is difficult. Tar formation depends on:

- Temperature level
- Type of biomass
- Moisture content
- Residence time in hot zones

Several researchers have focused on:

- Improving reactor throat design
- Introducing steam for reforming
- Using catalysts like dolomite or nickel

These improvements help further crack tar and make syngas cleaner for engine or burner applications.

Overall summary :

The existing literature shows that downdraft gasifiers are good for studying gas composition, temperature behaviour and tar formation, especially with uniform feedstocks like pine pillets. The system's natural ability to crack tar, along with stable burning characteristics of pellets, make it ideal for performance analysis and experimental validation.

Objectives of the study :

- This helps determine whether the downdraft gasifier can produce gas quality suitable for practical applications and how airflow influences CO, H₂, CH₄, and CO₂ concentrations
- To record detailed temperature values inside the gasifier we used five thermocouples placed at different heights corresponding to the drying, pyrolysis, oxidation, and reduction zones.
- To identify the precise temperature ranges where important gasification processes, like the start of pyrolysis, peak oxidation temperatures, and reduction reactions, take place
- To study how variations in temperatures and airflow affect tar formation and cracking within the gasifier
- To evaluate the overall performance of the downdraft gasifier using pine pellets, focussing on syngas quality and temperature stability , tar reduction efficiency under different conditions.

Work Plan:

STAGE1:

- Learning about how gasification works and how biomass turns into usable energy.
- Selecting the biomass and analyzing it.

STAGE2:

- Design and modify the parts that are required as a solution of the issue in the existing setup.
- Procuring the required materials.

STAGE3:

- Fabrication of the design parts.
- Assembling the fabricated materials.

STAGE4:

- Validate the constructed gasifier model by doing experiment.
- Record the results after observing the expected combustible producing gases.

Methodology

How much flowrate should be passed ?

- Typical Air flow rate sit around 1.3 -1.6 Nm³ per kg of biomass
- So, lets take around 1.5 Nm³ of air on average per kg
- Lets assume we took 2kg of biomass fuel = 2*1.5 Nm³/h = 3Nm³/h
- Now lets convert into L/min:

$$3*(1000/60) = 50\text{L/min}$$

Experimental setup :



Image 1.a

- This is our Biomass Gasifier setup , we modified the existing setup by remodelling the gasifier lid, to not let gases out from top of the lid.



Image 1.b -> Gasifier lid

- We installed a duct pipe with the exhaust fan in it . placed at the bottom of the Gasifier
- We did this modification , for not letting gases spread all over the place

How does this works ?

After loading the pine pellets into the biomass gasifier, we burnt the biomass with the help of camphor and a gas lighter . and then air passed through the inlet tube . While this is going on the gases are produced inside the gasifier . Now the exhaust fan will suck the gases from the gasifier .And the exhaust fan is attached in the duct pipe so those gases will flow through in the direction of how duct pipes are placed. We placed the duck pipes above the lab so no one will get diseased.



Image 1.c

- We modified the Air inlet tube to spread the gas evenly and more efficiently



Image 1.d -> Air inlet tube

- We built an ESP8266-based Wireless Temperature Monitoring Module with K-type Thermocouple MAX6675 Interface Board.

Why did we built this ?

- Real-time Temperature Monitoring:
It reads temperatures from multiple K-type thermocouples using MAX6675 converters and sends the data wirelessly to a laptop or mobile.
- Data Logging for Experiments:
It helps track temperature variations inside the gasifier (t1–t5 zones) during experiments for performance analysis and comparison.

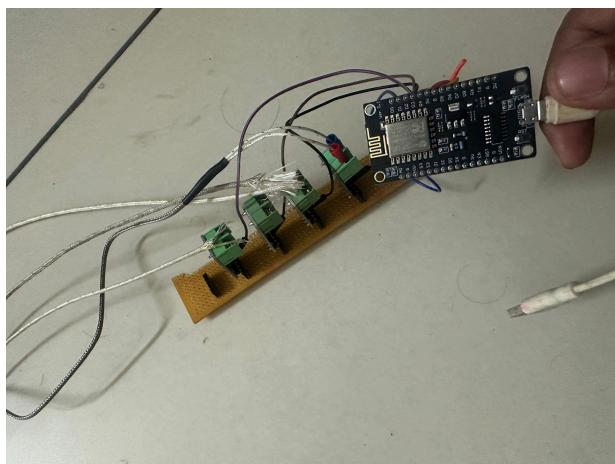


Image 1.e

Now we will see How we did the Gasification process :

- At first , we weigh the biomass and placed inside the biomass and then we burn the biomass fuel with the help of camphor and Gas lighter.



Weighing Biomass fuel



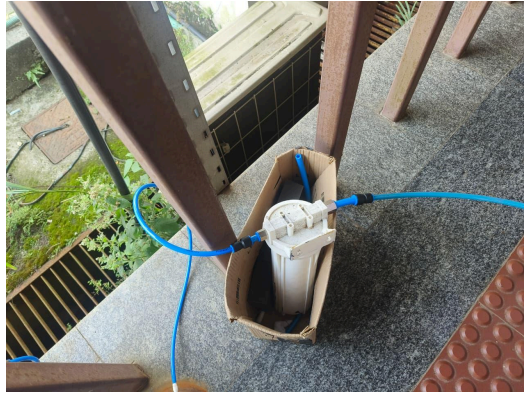
Burning the Biomass fuel

- We also inserted 5 thermocouples inside the Gasifier through drilled holes. These will help us to measure the temperature inside the Gasifier by ESP-8266 or Temperature analyzer .



K-type thermocouples inserted in Biomass Gasifier

- Before letting air pass, we will check if any leakage is present in the tubes with the help of soap water. If there are no leakages We will pass the air through air filter. Because a lot of soot particles and water vapor is accumulated in the exhaust pipe which can affect the gas analyzer so, we design the filter which is basically a PET pipe which has stack of cotton and metal scrub inside.



Air filter

- Now we will pass the Air through air inlet tube with certain flow rate for certain amount of biomass . Now the gases are taken by the metal probe . It is placed at the bottom of the gasifier and now the concentration of gases are measured by the Continuous Emission Monitoring system(Gas analyzer) .



Syngas Analyzer

Results and Discussions

Experiment 1

Operating Conditions

- Initial biomass: 1 kg pine pellets
- Initial airflow: 50 L/min at 2:50 PM
- Reduced airflow: 25 L/min at 2:58 PM
- Additional biomass: +1 kg added at 3:07 PM (because the first batch fully burned)
- Ignition from bottom: Started again with a slow 50 L/min airflow at 3:12 PM
- Experiment continued till the biomass completely burned off

Gas Concentration Behavior

CO (Carbon Monoxide)

- CO increases slowly during the first stable phase, then rises sharply after adding the second batch of biomass
- Peaks appear during fresh ignition and when oxidation is strong
- CO values correlate with high-temperature zones, showing good gasification activity

CO₂ (Carbon Dioxide)

- CO₂ rises early when biomass begins to burn cleanly
- Drops slightly when airflow decreases (less oxidation)
- After adding the second batch, CO₂ rises again due to renewed combustion

H₂ (Hydrogen)

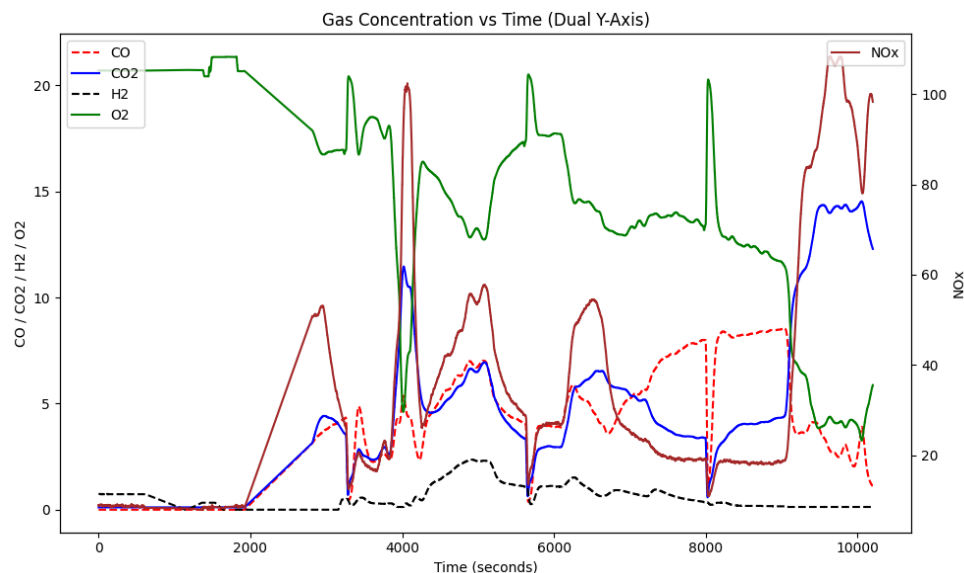
- Hydrogen remains low initially
- Increases when the reduction zone becomes active (after 3:07 PM)
- Maximum values seen during the most stable period after bottom ignition

O₂ (Oxygen)

- O₂ decreases when the biomass starts consuming oxygen properly
- Spikes indicate air imbalance or fresh ignition
- Drops again after 3:12 PM when strong reduction establishes

NO_x

- Mostly stable but rises sharply when temperatures suddenly increase
- Higher NO_x peaks correspond to high-temperature oxidation bursts

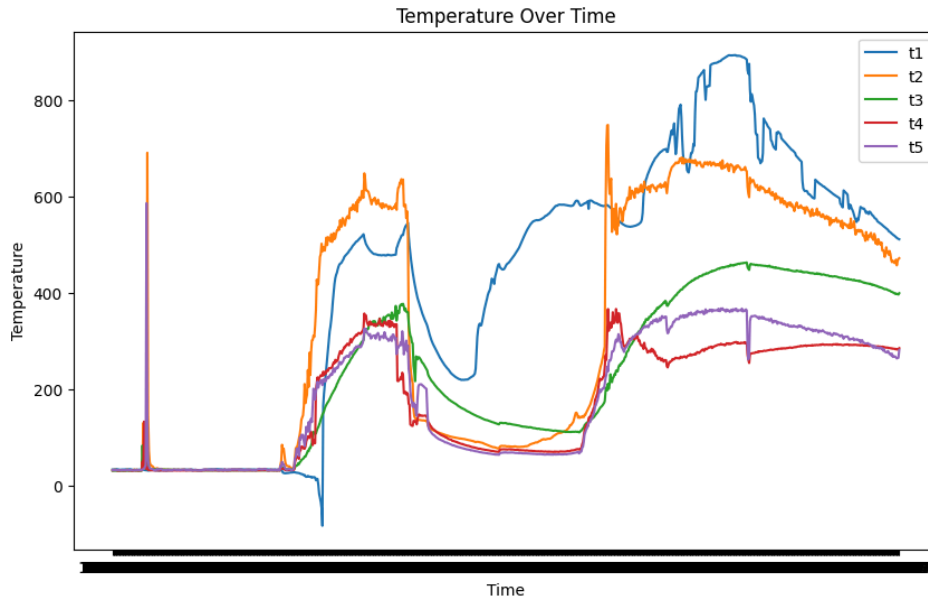


Temperature Behavior

- **Zone t5 (Upper):**
Slow rise, then stabilizes; peaks above 850 °C after second ignition.
- **Zone t4 (Upper-mid):**
Rises sharply after airflow change and fuel addition; reaches ~650–700 °C.
- **Zone t3 (Middle):**
Smooth increase to ~450 °C, showing steady pyrolysis/reduction.
- **Zone t2 (Lower-mid):**
Moderate rise, follows t3; dips during fuel drop, rises again after re-ignition.

➤ **Zone t1 (Bottom):**

Low at start, rises strongly after bottom ignition at 3:12; stays hot due to char oxidation.



➤ **Observation from the experiment 1**

- Temperature was low at the beginning with almost no gas formation.
- When airflow increased to 50 L/min, temperature rose fast and ignition started.
- CO₂ increased first, showing proper burning.
- CO and H₂ increased later, confirming gasification.
- O₂ dropped because it was fully consumed.
- Adding biomass at 3:12 caused sudden changes in temperature and gas levels.
- After 3:12, the CO, CO₂, and H₂ values fall within the global syngas range (CO: 15–25%, CO₂: 10–15%, H₂: 8–12%), proving good gasifier performance.
- Temperature zones showed a clear oxidation → reduction pattern, matching typical downdraft gasifier behavior.
- Gas trends remained stable toward the end, indicating steady-state gasification was achieved.

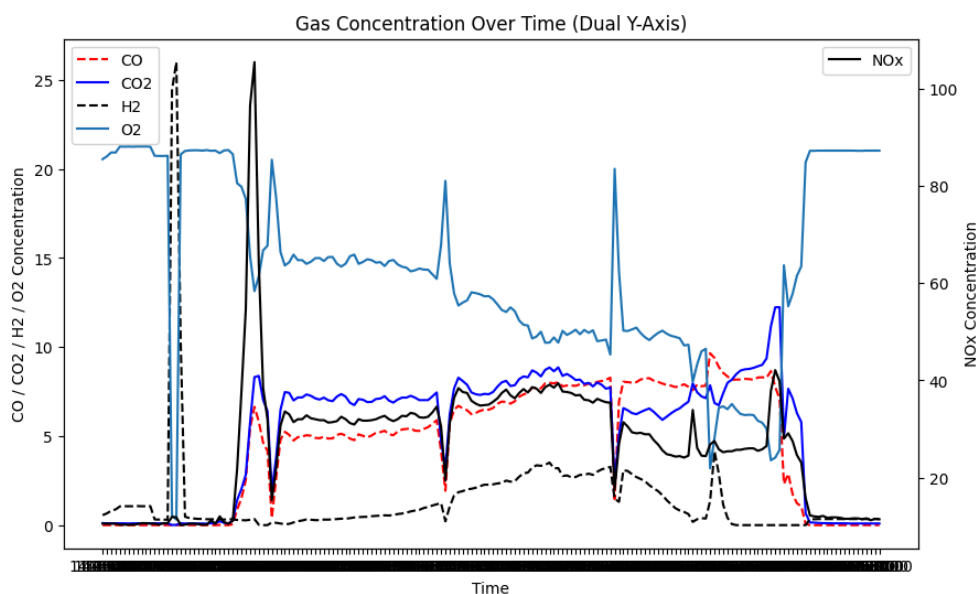
Experiment 2

Operating Conditions

- Initial biomass: 1.5 kg pine pellets
- Initial airflow: 20 L/min at 2:30 PM
- Ignition from the top : experiment started at 2:32 PM

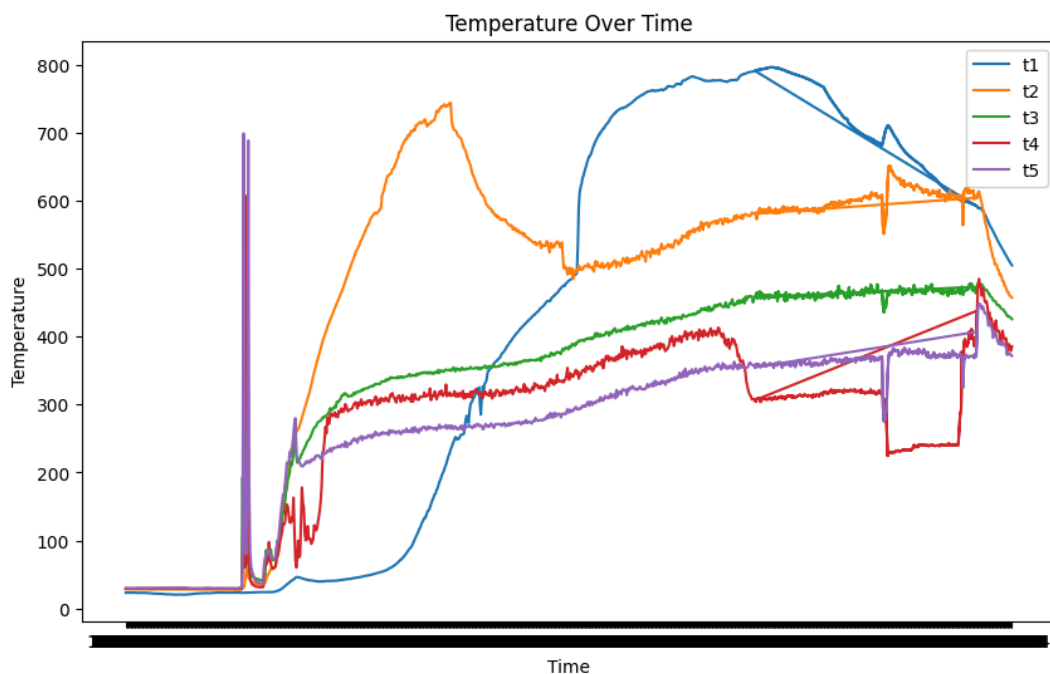
Gas concentration Behaviour

- CO₂ rapidly rose during ignition, which is normal for early strong combustion.
- Once steady gasification began, CO stabilized around typical syngas values (4–8%), indicating proper reduction reactions.
- CO₂ maintained a moderate range (5–10%), consistent with balanced oxidation.
- H₂ gradually increased toward mid-experiment, reaching ~2–3%, showing good pyrolysis and char reactions.
- O₂ levels dropped significantly during steady operation, confirming complete air utilization inside the reactor.
- NO_x showed short spikes during ignition and transition but remained stable afterward, indicating controlled combustion.
- The gas composition pattern matches expected downdraft gasifier behavior with clean producer gas generation.



Temperature Behaviour

- At the start, all five thermocouples (t1 bottom → t5 top) remain around 30–40°C, showing no reaction zone formation yet.
- After ignition from the top, t5 (top) is the first to rise, confirming flame propagation at the feedstock surface.
- As the combustion layer begins moving downward, t4 and t3 gradually rise, indicating the start of pyrolysis and early oxidation zones.
- t2 (near oxidation zone) reaches the highest temperature peak (~700–750°C). This shows the main oxidation zone is strongly active in this region.
- t1 (bottom) rises slower and remains lower initially, showing the reduction zone is the last to heat up.



- A large temperature spike is seen briefly across sensors (instrument glitch), but afterwards temperatures stabilize normally.
- After full stabilization, t2 stays between 550–600°C, while t3 and t4 stabilize between 350–450°C, indicating sustained burning.
- t1 eventually increases to 700–800°C, signaling that the oxidation front reached near the bottom later in the run.
- During the steady phase, temperature layering is correct:
 $t1 > t2 > t3 > t4 > t5$ matching typical downdraft behaviour.
- Towards the end, all temperatures slowly decrease as fuel is consumed and the combustion front weakens.

Overall observation from experiment 2

- The top ignition allowed heat to travel downward smoothly through the reactor.
- T5 and T2 reached the highest peaks early, confirming strong top oxidation.
- The temperature layers formed correctly, showing good downdraft function.
- The reactor remained stable with minimal temperature shocks.
- Gas composition followed expected behaviour once temperatures stabilized.
- Syngas formation improved as the char bed developed.
- CO and CO₂ stabilised at realistic downdraft values.
- H₂ remained moderate but consistent with semi-dry biomass.
- NO_x peaks matched temperature spikes in oxidation-heavy zones.
- Overall, Experiment 2 produced stable temperatures and steady syngas, showing efficient top-ignition performance.

Ignition at bottom had better results

Temperature Behaviour Comparison

Bottom Ignition – Exp-1

- Temperature rises quickly at t1 (bottom zone) → strong oxidation zone immediately.
- High peak temperatures (700–820°C) formed earlier and sustained longer.
- Stable heat transfer upward → t2, t3, t4, t5 rise gradually and consistently.
- No sudden collapses in temperature except during biomass recharge.
- Indicates good tar cracking, strong oxidation, faster syngas generation.

Top Ignition – Exp-2

- Temperatures rise slowly at the bottom (t1) because heat must travel downward.
- Middle and top zones (t4, t5) heat sooner than bottom → delayed reduction zone.
- Temperature peaks are lower and less stable than Exp-1

- Zones took longer to align into steady-state gasification mode
- Overall → slower to reach proper oxidation & reduction temperatures

Gas Analyzer Performance Comparison

Bottom Ignition – Exp-1

- CO and H₂ levels are more stable and higher for longer duration.
- CO₂ maintains a smoother trend.
- O₂ drops properly, indicating good combustion + reduction.
- NO_x shows controlled patterns → stable thermal reactions.

Top Ignition – Exp-2

- CO, H₂, and CO₂ take a longer time to stabilize.
- O₂ remains higher in many regions → incomplete conversion early in the run.
- More fluctuations in CO and NO_x → less stable reaction front

Final conclusion

Bottom Ignition (Exp-1) is clearly better because

- It forms the oxidation and reduction zones faster.
- Produces higher and more stable syngas (CO + H₂).
- Achieves higher temperatures, meaning better tar cracking.
- O₂ drops properly → gasification is complete and efficient.
- The entire temperature profile behaves exactly like an ideal downdraft gasifier.

Top ignition(Exp -2) is slower, inefficient at the start, and gives unstable gas composition until temperatures equalize.

Bibliography

- Reed, T. B., & Das, A. *Handbook of Biomass Downdraft Gasifier Engine Systems*. Solar Energy Research Institute, 1988.
- Basu, P. *Biomass Gasification, Pyrolysis and Torrefaction*. Academic Press, 2018.
- Knoef, H. A. M. (Ed.). *Handbook Biomass Gasification*. BTG Biomass Technology Group, 2005.
- Erlich, C., & Fransson, T. H. "Downdraft gasification of biomass: Temperature distribution and gas composition." *Fuel*, 84, 2005.
- AI Assistance: ChatGPT (OpenAI). Used for organizing observations, simplifying explanations, and comparing syngas values with global ranges.